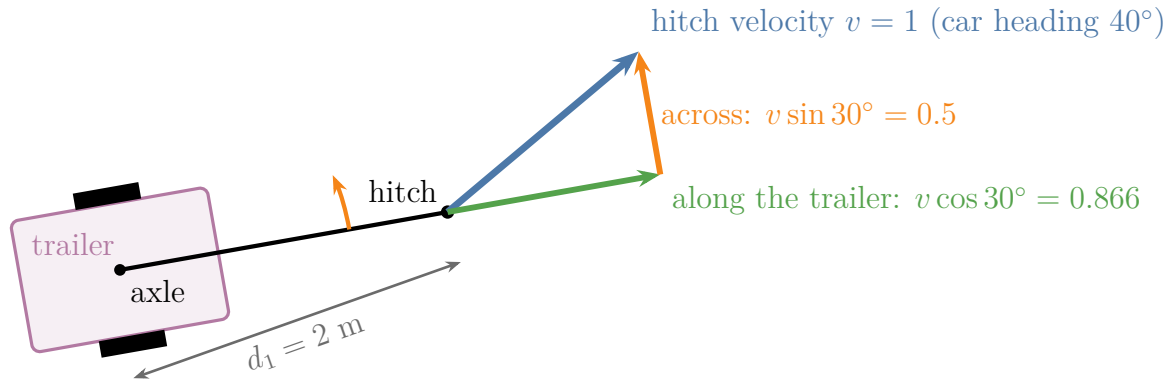




the axle cannot slide sideways, so the across part swings the trailer about its axle:

$$\text{turn rate} = \text{across speed} \div \text{distance} = 0.5 \div 2 = 0.25 \text{ rad/s}$$